

Geometry-Topology Qualifying Examination — September 25, 2002

Problem 1

Suppose $P(x, y, z)$, $Q(x, y, z)$, and $R(x, y, z)$ are C^∞ functions on $\mathbb{R}^3$ which vanish identically if $|x| \geq 5$, $|y| \geq 5$, or $|z| \geq 5$. Prove that the volume integral

$$\int_{-6}^{+6} \int_{-6}^{+6} \int_{-6}^{+6} d(Pdy \wedge dz + Qdx \wedge dz + Rdx \wedge dy) = 0.$$

(Do this directly, not by quoting Stokes' Theorem: this is a special case of the proof of Stokes' Theorem!)

Solution.

$$d(Pdy \wedge dz + Qdx \wedge dz + Rdx \wedge dy) = P_x dx \wedge dy \wedge dz + Q_y dy \wedge dx \wedge dz + R_z dz \wedge dx \wedge dy$$

$$\int_{[0,6]^2} \int_{-6}^6 P_x dx \wedge dy \wedge dz = \int_{-6}^6 (P(6) - P(-6)) dy \wedge dz = 0.$$

Similarly for Q and R . Thus the total integral is 0.

Problem 2 [TODO]

Suppose that $V = P(x, y, z) \frac{\partial}{\partial x} + Q(x, y, z) \frac{\partial}{\partial y} + R(x, y, z) \frac{\partial}{\partial z}$ is a C^∞ vector field on $\mathbb{R}^3$ with $V \neq \vec{0}$ at the origin. Find a necessary and sufficient condition for there to exist a C^∞ function $\lambda(x, y, z)$ in some neighborhood of the origin such that λV is the gradient of a C^∞ function on the neighborhood.

Proof:

□

Problem 3

Let $T_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the right-hand rule rotation around the positive z -axis by t degrees and $S_s : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the right-hand-rule rotation around the positive x -axis by t degrees.

- Find the infinitesimal generators of the flows T_t and S_t , i.e., the vector fields X and Y , respectively, on $\mathbb{R}^3$ whose flows are $\{T_t\}$ and $\{S_t\}$.
- Compute the commutator $T_{-t} \circ S_{-t} \circ T_t \circ S_t$.
- Compare the result of (b) (lowest order non-identically zero term) with the Lie bracket $[X, Y]$.

Proof: We have that:

$$T_t(\vec{x}) = \begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \vec{x}$$

$$S_t(\vec{x}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(t) & -\sin(t) \\ 0 & \sin(t) & \cos(t) \end{pmatrix} \vec{x}$$

1.

$$\begin{aligned} X(\vec{x}_0) &= \frac{\partial}{\partial t} [T_t(x)]_{t=0} \\ &= \frac{\partial}{\partial t} \left[\begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \vec{x} \right]_{t=0} (x) \\ &= \begin{pmatrix} -\sin(t) & -\cos(t) & 0 \\ \cos(t) & -\sin(t) & 0 \\ 0 & 0 & 0 \end{pmatrix}_{t=0} \vec{x} \\ &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A\vec{x} \\ &= -y\partial_x + x\partial_y \\ Y(\vec{x}_0) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \vec{x} = B\vec{y} = -z\partial_y + y\partial_z \end{aligned}$$

2. We have

$$\begin{aligned} T_t &= I + At + \frac{1}{2}t^2A + O(t^3) \\ S_t &= I + Bt + \frac{1}{2}t^2B + O(t^3) \\ T_{-t} &= I - At + \frac{1}{2}t^2A + O(t^3) \\ S_{-t} &= I - Bt + \frac{1}{2}t^2B + O(t^3) \end{aligned}$$

Thus

$$T_{-t}S_{-t}T_tS_t = \left(I - At + \frac{1}{2}t^2A + O(t^3) \right) \left(I - Bt + \frac{1}{2}t^2B \right)$$

□

Problem 4

Take as given that a C^∞ 2-form ω on S^2 is of the form $d\theta$ for some C^∞ 1-form θ if and only if $\int_{S^2} \omega = 0$. Use this to show that every C^∞ 2-form Ω on $\mathbb{R}P^2$ has the form $d\Lambda$ for some C^∞ 1-form Λ . (Do not just quote DeRham's Theorem here.)

Proof: Let A be the antipodal map of S^2 . $q : S^2 \rightarrow \mathbb{R}P^2$ standard quotient map, which induces $q^* : \Omega^2(\mathbb{R}P^2) \rightarrow \Omega^2(S^2)$. Note that q^* is an injective map. Note $A^*(q^*(\omega)) = q^*(\omega)$. But note A flips the orientation of S^2 . $\forall \psi \in \Omega^2(\mathbb{R}P^2)$,

$$\begin{aligned} I &= \int_{S^2} q^*(\psi) = \int_{S^2} A^*(q^*(\psi)) \\ &= \int_{A(S^2)} q^*(\psi) = - \int_{S^2} q^*(\psi) \Rightarrow I = 0 \end{aligned}$$

Thus $q^*(\psi) = d\theta$. $d\theta = A^*(q^*(\psi)) = A^*(d\theta) = d(A^*(\theta))$. Let $\tilde{\theta} = \frac{1}{2}[\theta + A^*(\theta)]$, note $A^*(\tilde{\theta}) = \tilde{\theta}$. Thus $\tilde{\theta}$ factors through $\Omega^1(\mathbb{R}P^2)$. Thus $\psi = d\tilde{\theta}$. □

Problem 5

- (a) Suppose $F : S^1 \rightarrow \mathbb{R}^3$ is a C^∞ function such that dF is nowhere zero (on S^1). Prove that there is a two-dimensional subspace P of $\mathbb{R}^3$ such that $\pi_P \circ F : S^1 \rightarrow \mathbb{R}^3$ has nowhere vanishing differential, where π_P = orthogonal projection on P .
- (b) Show by example (a picture with explanation is all right) that there is such an F that is also 1 to 1 (injective) but is such that, for all P , $\pi_P \circ F$ fails to be injective.
- (c) Show that if $F : S^1 \rightarrow \mathbb{R}^4$ is C^∞ and injective then there is a three-dimensional subspace H of $\mathbb{R}^4$ such that $\pi_H \circ F$ is injective, where π_H = orthogonal projection on H .

Proof: (a): Since $dF_p \neq 0$ for all $p \in S^1$. Thus dF_p is an injective map. Thus F is an immersion. □

Problem 6

- (a) Suppose $F : S^n \rightarrow S^n$ is fixed-point free (i.e., for all $p \in S^n$, $p \neq F(p)$). Show that F is homotopic to the antipodal map $p \mapsto -p$, $p \in S^n$.
- (b) Use part (a) to show that every vector field on (tangent to) S^{2n} , $n = 1, 2, 3, \dots$, vanishes somewhere on S^{2n} (i.e., has a zero).

Proof: (a): Suppose $F : S^n \rightarrow S^n$ is fixed point free. Let $H : S^n \times I \rightarrow S^n$ where

$$H(x, t) = \frac{F(x)t - (1-t)x}{\|F(x)t - (1-t)x\|}$$

Note since $F(x) \neq x$ for all $x \in S^1$, $\nexists t \in [0, 1]$ such that $F(x)t - (1-t)x = 0$. Thus H is a homotopy between $x \rightarrow -x$ and F .

(b): Let X be a vector field. Assume the contrary, $\forall p \in S^{2n}$ we have that $X(p) \neq p$. Since S^{2n} is compact and $\|X(\cdot)\|$ is smooth, $\exists M \in \mathbb{R}$ such that $\|X\|_\infty < M$. Let $\varepsilon > 0$ be small enough such that $\varepsilon M < \pi$. Let $F : S^{2n} \times [0, 1] \rightarrow S^{2n}$ where $F(p, t) = \exp_p(t\varepsilon X(p))$. Thus F is a homotopy between the identity map and $F(\cdot, 1)$ a map with no fixed point which is homotopic to antipodal map. $\deg(\text{id}_{S^{2n}}) = 1 = \deg(F(\cdot, 1)) = \deg([x \rightarrow -x]) = (-1)^{2n-1} = -1$. This is a contradiction. □

Problem 7 [TODO]

- (a) Discuss carefully how to obtain the long exact sequence in homology from a short exact sequence of chain complexes. (Include definitions of the maps in the long exact sequence.)
 (b) If the short exact sequence is

$$0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow 0,$$

prove exactness of the long exact sequence at $H_k(C_3)$ [in $\dots H_k(C_2) \rightarrow H_k(C_3) \rightarrow H_{k-1}(C_1) \dots$].

Problem 8

- (a) Suppose $F : T^2 \rightarrow T^2$ (where $T^2 = S^1 \times S^1$) is a continuous function such that $F(p) = p$ for some $p \in T^2$ and

$$F_* : \pi_1(T^2, p) \rightarrow \pi_1(T^2, p)$$

is the identity map. Is F necessarily homotopic to the identity map from T^2 to itself?

- (b) Is a C^∞ map $F : T^2 \rightarrow T^2$ of degree 1 necessarily homotopic to the identity map of T^2 to itself? Explain/prove your answer. [TODO]

Solution. (a): $\tilde{F} : T^2 \rightarrow \mathbb{R}^2$ where $\tilde{F}(x, y) = F(x, y) + (1, 1)$. Note $F = q \circ \tilde{F}$. Let H be the standard straight line homotopy from $\text{id} \rightarrow \tilde{F}$. Thus $q \circ H$ is the straight line homotopy from $\text{id} \rightarrow F$.

(b):

Problem 9

- (a) Discuss the representation of $\mathbb{C}P^n$ as a cell complex.
 (b) Use part (a) to find the homology of $\mathbb{C}P^n$: prove carefully that your calculation is correct.

Solution. (a): You have a 0-cell, 2-cell, ..., $2n$ -cell. To attach the cells, you attach cell $2(n-1)$ to cell $2n$ by attaching the boundary to $2(n-1)$ cell directly.

(b): You have the complex neighborhood of s :

$$0 \rightarrow C_{2n} = \mathbb{Z} \rightarrow 0 \rightarrow C_{2(n-1)} = \mathbb{Z} \rightarrow \dots \xrightarrow{d_3} C_2 = \mathbb{Z} \xrightarrow{d_2} 0 \xrightarrow{d_1} C_0 = \mathbb{Z} \rightarrow 0$$

Thus we have $0 \xrightarrow{d_{2k}} C_{2k} \xrightarrow{d_{2k-1}} 0 \Rightarrow H_{2k}(\mathbb{C}P^n) = \mathbb{Z}$ for $H_{2k-1}(\mathbb{C}P^n) = 0$.

Problem 10

- (a) Let X be the space obtained by attaching two discs to S^1 , the first disc being attached by $S^1 = \partial D_1 \rightarrow S^1$ being the 7 times around (counterclockwise) map, e.g., $z \rightarrow z^7$, $|z| = 1$, $z \in \mathbb{C}$, and the second being attached by $S^1 = \partial D_2 \rightarrow S^1$ being the 5 times around map $z \rightarrow z^5$. Find the homology of X .
 (b) Can X be made a C^∞ manifold? Why or why not?

Proof: (a): We have this cellular chain complex:

$$0 \rightarrow \mathbb{Z}^2 \xrightarrow{\begin{pmatrix} 7 & 5 \end{pmatrix}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0.$$

Thus

$$H_0(X) = \mathbb{Z}$$

$$H_1(X) = \frac{\mathbb{Z}}{7\mathbb{Z} + 5\mathbb{Z}} = 0$$

$$H_2(X) = \mathbb{Z} \left\langle \begin{pmatrix} -5 \\ 7 \end{pmatrix} \right\rangle = \mathbb{Z}$$

$$7x + 5y = 0 \Rightarrow 7x = -5y \Rightarrow x = -\frac{5}{7}y$$

(b): X consists of finitely many cells, and hence is compact. Furthermore it has a dimension $d = 2$. Assume the contrary, that X is a 2 dimensional C^∞ manifold. Fix a $p \in S^1 \subset X$. There exists a neighborhood of p , N such that N is diffeomorphic to U in $\mathbb{R}^2$. Because the attaching maps of the disks are $z \rightarrow z^7$ and $z \rightarrow z^5$ respectively, p has exactly 7 preimages on the boundary of disk 1 and 5 preimages on the boundary of disk 2. Therefore the neighborhood N contains an open interval $I \subset S^1$, where the 7 half disks from the first disk and the 5 half disks from the 2nd disk are all glued together at the interval I . Now if we look at $N \setminus I$, it has 12 connected components. Whereas $U \setminus I$, a arc in $\mathbb{R}^2$, has exactly 2 connected components. Thus U and N cannot be homeomorphic hence diffeomorphic. Thus a contradiction occurs and X cannot be a C^∞ manifold.

□

Reflection

I want to review the following topics:

1. Local Degree (Poincare Duality in Hatcher + Differential topology). Broadly degree theory - Hatcher Problems + Differential Topology Perspective.
2. Cellular Homology
3. Integrability and Frobenius Theorem